2. Elimination reactions of alcohols

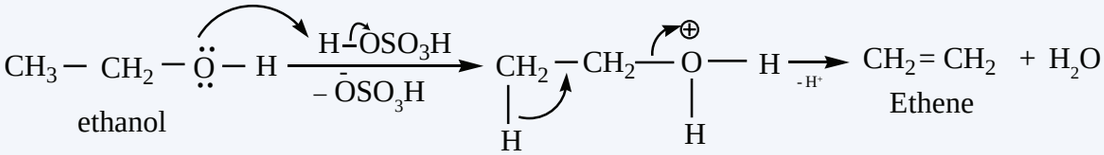
When alcohols are heated with a suitable dehydrating agents like sulphuric acid, the H and OH present in the adjacent carbons of alcohols are lost, and it results in the formation of a carbon – carbon double bond. Phosphoric acid, anhydrous $ZnCl_2$, alumina etc., can also be used as dehydrating agents.

Example

$$CH_3 - CH_2 - OH \xrightarrow[443\ K]{H_2SO_4} CH_2 = CH_2 + H_2O$$

Mechanism

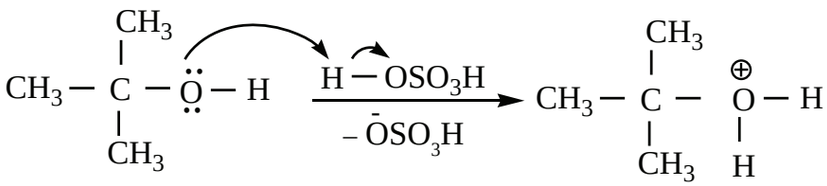
Primary alcohols undergo dehydration by E_2 mechanism



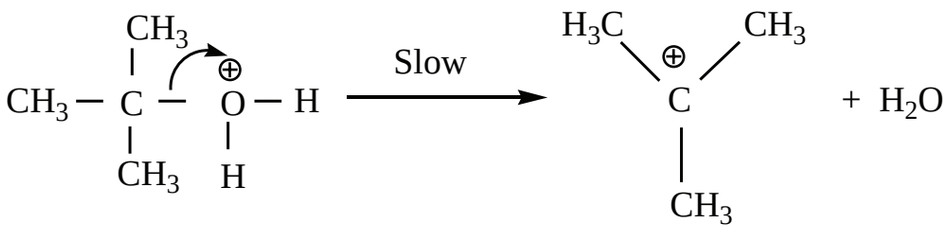
Tertiary alcohols undergo dehydration by E_1 mechanism. It involves the formation of a carbocation.

Protonation of alcohol

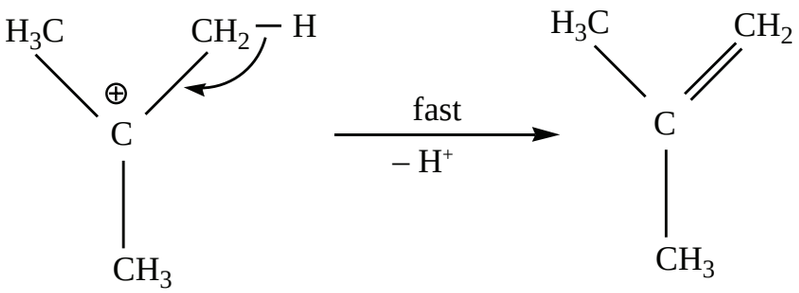
Step 1 :



Step 2 : Dissociation of oxonium ion to form a carbocation.



Step 3 : Deprotonation of carbocation to form an alkene



2 - methylprop - 1- ene

Order of reactivity:

The relative reactivities of alcohols in the dehydration reaction follows the order

primary < secondary < tertiary

XII U11-Hydroxy compounds.indd 115 11-01-2021 15:44:48